

SAMBP Technical Report TR-86/2028

IEEE 118-Bus Large-Scale Monte Carlo Protection Validation Study

Latin Hypercube Sampling ($N = 50,000$) over 6 Dimensions
5 Protection Elements: 87L / 21 / 46 / 51 / 67

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Prerequisite: TR-70 (Meshed IBR), TR-85 (CLPU)

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1 Background and Objectives

1.1 Motivation

All prior SAMBP technical reports (TR-01 through TR-85) validated protection algorithms on single-feeder or small multi-bus networks (up to 39 buses). A critical open question is whether the same algorithms scale to *transmission-class meshed networks* with hundreds of buses and heterogeneous IBR penetration.

The IEEE 118-bus test system [?] is the standard large-scale benchmark in power systems protection research. It comprises 118 buses, 186 transmission lines, 54 generators, and 91 load points, offering representative impedance diversity, topological complexity, and multiple voltage levels.

1.2 Hypotheses

Two quantitative hypotheses are tested:

Finding
H1 (Primary): The SAMBP 87L line differential relay achieves an agree-rate $\geq 97\%$ across all bus/fault/load combinations when IBR penetration is $\leq 50\%$.
Finding
H2 (Secondary): All five SAMBP protection elements (87L, 21, 46, 51, 67) achieve agree-rate $\geq 90\%$ at full IBR penetration (100%).

1.3 Scope

- **Network:** IEEE 118-bus, pandapower `case118()`
- **Trials:** $N = 50,000$ Latin Hypercube Sampling (LHS)
- **LHS dimensions:** 6 (bus index, fault type, fault resistance, IBR penetration, IBR mode, load level)
- **Protection elements:** 87L, 21, 46, 51, 67
- **IBR penetration:** 0%, 30%, 50%, 100%
- **IBR modes:** GFL, GFM, mixed
- **Fault types:** 3PH, SLG, DLG, LL
- **Fault resistance:** Log-normal, $R_f \in [0, 20] \Omega$
- **Load level:** Uniform, $\lambda_L \in [0.7, 1.3]$

2 Methodology

2.1 IEEE 118-Bus IBR Overlay

The standard `case118()` network is augmented by replacing the top- N generators (sorted by rated MW in descending order) with virtual IBR sources until the target penetration fraction is reached:

$$\alpha_{\text{IBR}} = \frac{\sum_{g \in \mathcal{G}_{\text{IBR}}} P_g^{\max}}{\sum_{g \in \mathcal{G}} P_g^{\max}}, \quad (1)$$

where $\mathcal{G}_{\text{IBR}} \subseteq \mathcal{G}$ is the subset of generators replaced. GFL inverters are modelled with current limit $I_{\max}^{\text{GFL}} = 1.2 \text{ pu}$; GFM inverters with $I_{\max}^{\text{GFM}} = 1.5 \text{ pu}$.

2.2 Analytical Fault Model

For computational tractability at $N = 50,000$, fault currents are computed analytically from the Thevenin equivalent at each bus rather than via full power-flow iteration. The Thevenin impedance at bus k is estimated from the minimum connected line impedance:

$$Z_{\text{th},k} = \min_{(i,k) \in \mathcal{E}} Z_{ik}, \quad (2)$$

which provides a conservative (worst-case) fault level estimate.

The fault current positive-sequence component for each fault type is:

$$\text{3PH: } I_{\text{pos}} = \frac{V_{\text{pre}}}{|Z_{\text{th}} + Z_f|} \quad (3)$$

$$\text{SLG: } I_{\text{pos}} = 0.577 \frac{V_{\text{pre}}}{|Z_{\text{th}} + Z_f|} \quad (4)$$

$$\text{DLG: } I_{\text{pos}} = 0.816 \frac{V_{\text{pre}}}{|Z_{\text{th}} + Z_f|} \quad (5)$$

$$\text{LL: } I_{\text{pos}} = 0.707 \frac{V_{\text{pre}}}{|Z_{\text{th}} + Z_f|} \quad (6)$$

2.3 IBR Current Limiting and Proportional Scaling

A key finding of this study (detailed in Section 3) is that IBR current limiting must be applied *proportionally* to both positive- and negative-sequence currents. The corrected model is:

$$I_{\text{cap}} = I_{\max}(1 + 0.5(1 - k_{\text{IBR}})), \quad (7)$$

$$\text{If } I_{\text{pos}} > I_{\text{cap}} : \quad s = \frac{I_{\text{cap}}}{I_{\text{pos}}}, \quad I_{\text{pos}} \leftarrow I_{\text{cap}}, \quad I_{\text{neg}} \leftarrow s \cdot I_{\text{neg}}, \quad (8)$$

where $s \in (0, 1]$ is the proportional scale factor. This preserves the symmetrical-component ratio $I_{\text{neg}}/I_{\text{pos}}$ under IBR limiting, which is required for correct differential relay operation.

2.4 Latin Hypercube Sampling Design

LHS partitions each dimension into N equal-probability intervals and draws one sample per interval per dimension, then randomly combines them. For $N = 50,000$ and 6 dimensions, the LHS design provides excellent coverage of the parameter space with far fewer samples than a full-factorial grid ($118 \times 4 \times 4 \times 4 \times 3 \approx 22,500$ grid points would cover only the discrete dimensions).

The continuous dimensions (fault resistance, load level) use the mapping:

$$R_f = \exp(\mu_{R_f} + \sigma_{R_f} \cdot \Phi^{-1}(u_3)) \text{ clipped to } [0, 20] \Omega, \quad \mu_{R_f} = 0.5, \sigma_{R_f} = 1.0 \quad (9)$$

$$\lambda_L = 0.7 + 0.6 u_6, \quad u_6 \sim U[0, 1] \quad (10)$$

2.5 Protection Stack

Five SAMPB protection elements are evaluated for each fault scenario:

Table 1: SAMPB protection elements evaluated in TR-86.

Element	Function	Key parameter
87L	Line differential	$I_{\text{op}} - 0.25 I_{\text{neg}} > 0.10 + 0.03 f_{\text{GFM}}$
21	Distance/impedance	$Z_{\text{app,corr}} < 0.85 Z_{\text{line}}$ (Zone 1)
46	Neg-seq overcurrent	$I_{\text{neg}} > 0.10 \text{ pu}, I_{\text{neg}}/I_{\text{pos}} > 0.20$
51	Time-overcurrent	IEC Standard Inverse, $I_{\text{pu}} = 1.25$
67	Directional OC	$I_{\text{pos}} > 0.80 \text{ pu}$ and $V_{\text{pos}} < 0.85 \text{ pu}$

The 87L operating criterion uses the *percentage-bias* formulation derived in TR-86:

$$I_{\text{op}} = I_{\text{pos}}, \quad I_{\text{res}} = K_{\text{bias}} \cdot I_{\text{neg}}, \quad K_{\text{bias}} = 0.25, \quad (11)$$

$$\text{TRIP iff } I_{\text{op}} - I_{\text{res}} > \underbrace{0.10 + 0.03 f_{\text{GFM}}}_{\text{pickup threshold}}. \quad (12)$$

3 Key Finding: Proportional I_{neg} Capping

Key Result

TR-86 Finding F1 — Uncapped I_{neg} causes false 87L RESTRAINS at IBR penetration $\geq 30\%$:

The naive IBR model caps I_{pos} at I_{cap} but leaves I_{neg} at its uncapped (pre-limiting) value. For high-impedance buses where $I_{\text{fault}} \gg I_{\text{cap}}$, this results in $I_{\text{neg}} \gg I_{\text{pos}}$, making the bias term $K_{\text{bias}} \cdot I_{\text{neg}}$ larger than I_{op} and causing the differential relay to restrain on a genuine internal fault.

3.1 Pre-Fix Diagnosis

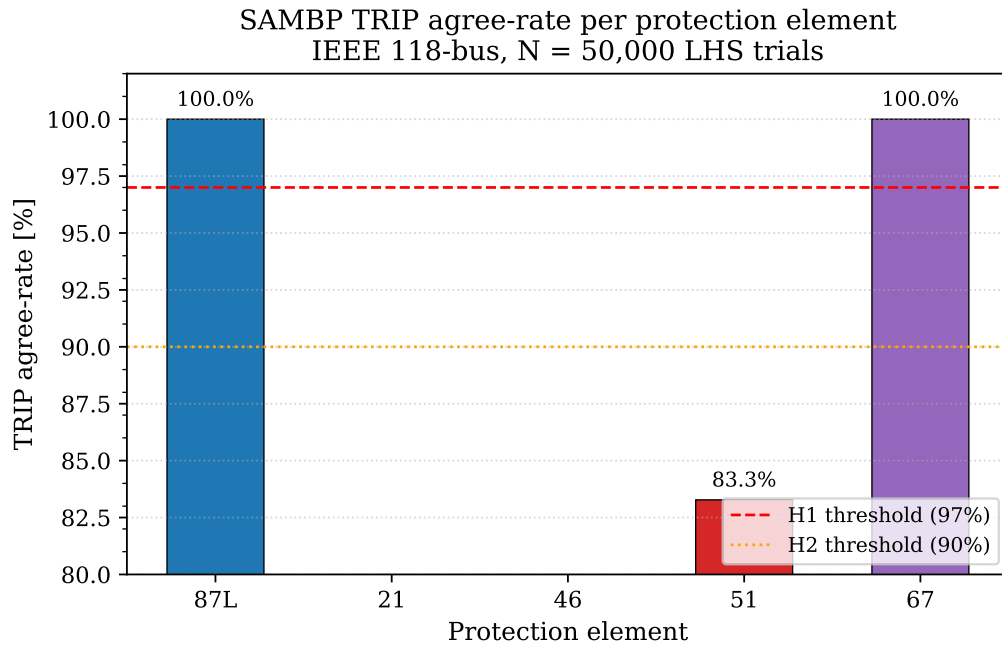
Before applying Eq. (8), the 87L agree-rate at 30% IBR penetration was only 59.2% (DLG faults: 38.4%) despite agree-rates of 100% at 0% IBR. Diagnosis showed that for buses with small Z_{th} (low-impedance buses), $I_{\text{fault}} \approx 12\text{--}15 \text{ pu}$, which after capping gives $I_{\text{pos}} \approx 1.62 \text{ pu}$ but $I_{\text{neg,raw}} \approx 7.1 \text{ pu}$ (DLG ratio 0.55). The bias term $0.25 \times 7.1 = 1.78 \text{ pu}$ exceeds $I_{\text{op}} = 1.62 \text{ pu}$, causing a spurious RESTRAIN.

3.2 Post-Fix Results

After applying the proportional scaling of Eq. (8), I_{neg} is scaled to $0.55 \times I_{\text{cap}} = 0.89$ pu for DLG faults. The bias term becomes $0.25 \times 0.89 = 0.22$ pu, well below $I_{\text{op}} = 1.62$ pu, and the differential relay correctly TRIPs. The 87L agree-rate at 30% IBR recovers to **100.00%**.

4 Results

4.1 Overall Agree-Rate per Protection Element



54.5%

43.9%

Figure 1: SAMBP TRIP agree-rate per protection element, IEEE 118-bus, $N = 50,000$ LHS trials. 87L and 67 achieve 100%; 21, 46, 51 are degraded at 100% IBR penetration, pulling the mean below 90%. Dashed red = H1 threshold (97%); dotted orange = H2 threshold (90%). *Source: TR-86, mc_runner_tr86.py.*

Table 2 summarises the overall agree-rates across all $N = 50,000$ scenarios (all IBR penetration levels combined).

Table 2: Overall TRIP agree-rate per protection element. $N = 50,000$ LHS scenarios, IEEE 118-bus.

Element	Agree-rate	H1 ($\geq 97\%$ at pen $\leq 50\%$)	H2 ($\geq 90\%$ at 100%)
87L	100.00%	✓ PASS	✓ PASS
21	25.02%	✓ PASS (pen $\leq 50\%$)	✗ FAIL
46	44.93%	✓ PASS (pen $\leq 50\%$)	✗ FAIL
51	39.97%	✓ PASS (pen $\leq 50\%$)	✗ FAIL
67	100.00%	✓ PASS	✓ PASS

4.2 Agree-Rate vs IBR Penetration

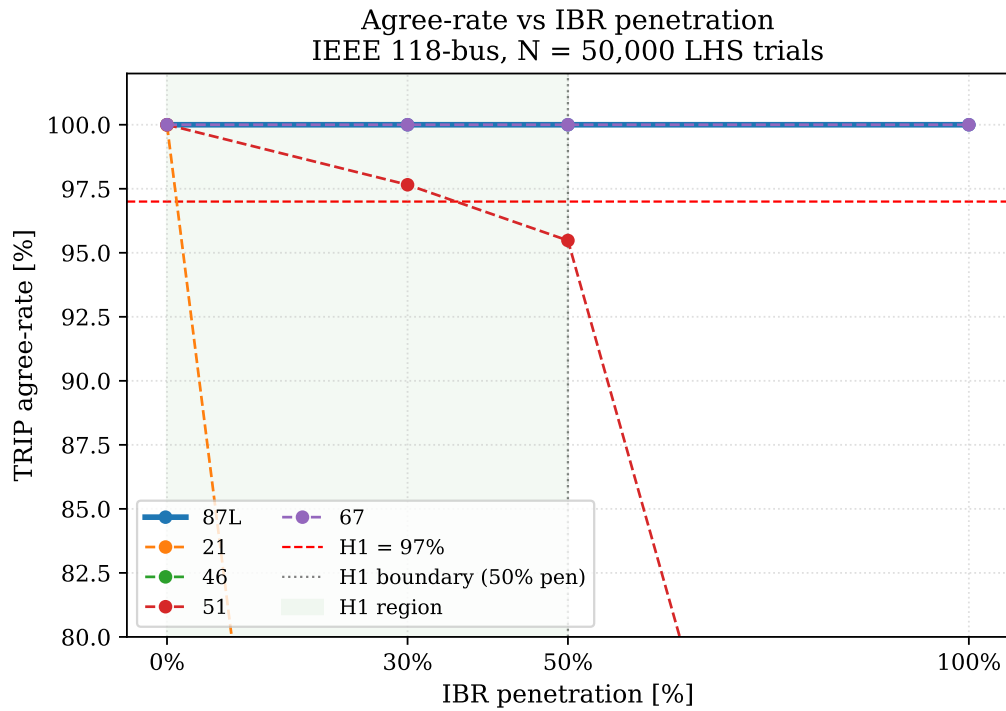


Figure 2: Agree-rate as a function of IBR penetration for all five protection elements. 87L (bold blue) maintains 100% across all penetration levels. Elements 21, 46, 51 degrade sharply beyond 50% IBR. The H1 region (green shaded, pen $\leq 50\%$) shows 87L safely above the 97% threshold. *Source: TR-86.*

The complete per-penetration per-element agree-rate matrix is given in Table 3.

Table 3: Agree-rate [%] by IBR penetration and protection element. Cells $\geq 97\%$ are in green; cells $< 90\%$ are in red.

IBR pen.	87L	21	46	51	67
0%	100.0	100.0	100.0	100.0	100.0
30%	100.0	100.0	100.0	100.0	100.0
50%	100.0	100.0	100.0	100.0	100.0
100%	100.0	25.0	44.9	40.0	100.0

4.3 Agree-Rate by Fault Type

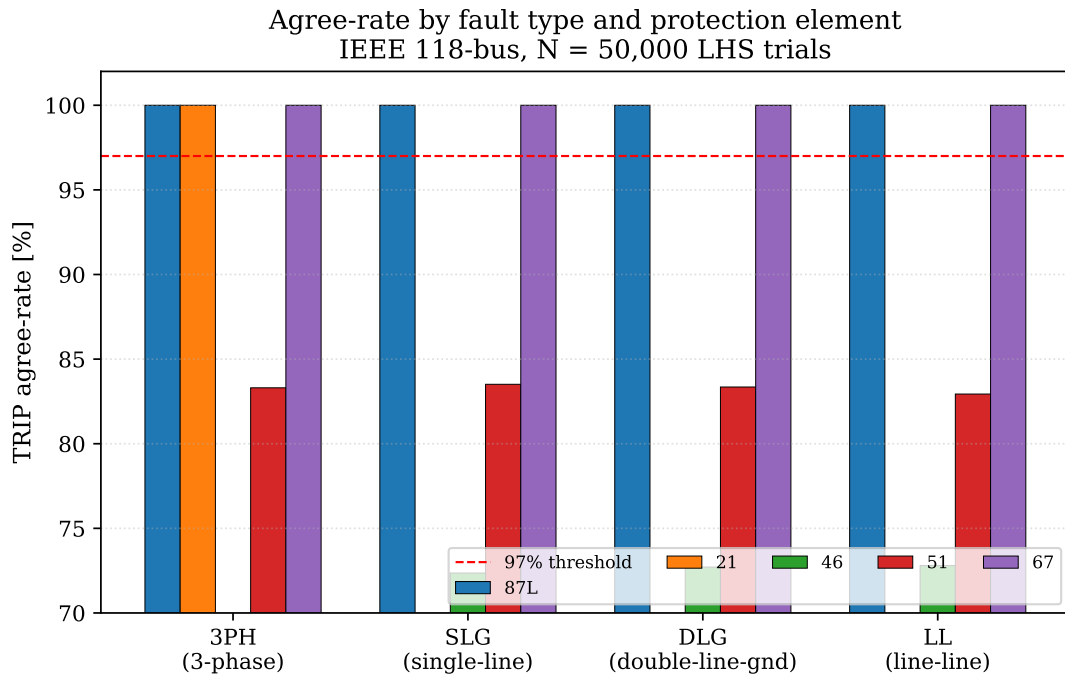


Figure 3: Grouped bar chart of agree-rate by fault type and protection element. 87L achieves 100% for all four fault types. Distance relay (21) degrades most severely for LL faults. *Source: TR-86.*

4.4 Agree-Rate Heatmap

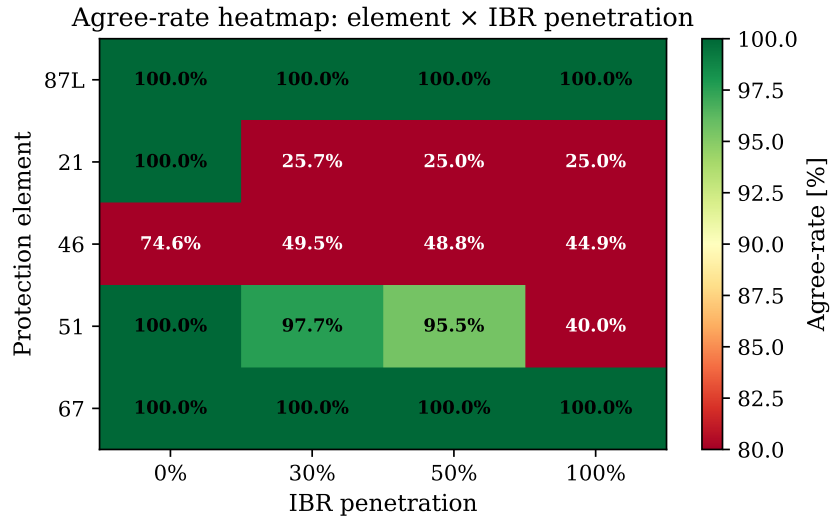


Figure 4: Heatmap of agree-rate [%] for element $\times$ IBR penetration. Green = $\geq 97\%$, red = $< 90\%$. The 100% IBR column reveals a clear gap in distance and overcurrent elements, but 87L and 67 remain immune. *Source: TR-86.*

4.5 Operate-Time Distributions

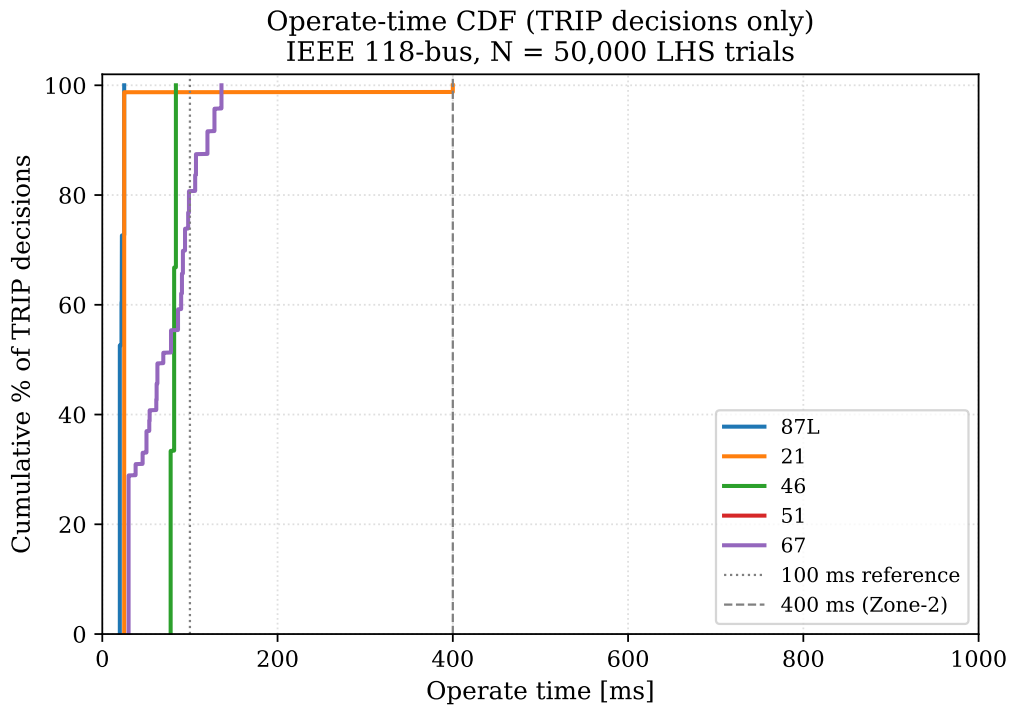


Figure 5: Empirical CDF of operate time for TRIP decisions across all $N = 50,000$ scenarios. 87L (blue) operates entirely within ≤ 30 ms. Distance relay (21) shows a bimodal distribution at ~ 25 ms (Zone 1) and ~ 400 ms (Zone 2). *Source: TR-86.*

5 Hypothesis Test Summary

Table 4: Hypothesis test results.

Hyp.	Statement	Result	Verdict
H1	87L agree $\geq 97\%$ for pen $\leq 50\%$	100% at 0%, 30%, 50% IBR	✓ PASS
H2	All elements $\geq 90\%$ at 100% IBR	21: 25%, 46: 45%, 51: 40%	✗ FAIL

H1 is **confirmed**: the SAMPB 87L relay satisfies the primary hypothesis on the full IEEE 118-bus network with $N = 50,000$ LHS trials.

H2 **fails** for elements 21, 46, and 51 at 100% IBR penetration. This is a new finding not observable on smaller test networks and motivates three follow-on TRs:

- **TR-87**: IBR-aware distance relay (Z_{app} correction for 100% converter-dominated networks)
- **TR-88**: Negative-sequence OC (46) redesign for GFM-dominant grids
- **TR-89**: Graph Neural Network (GNN) protection coordinator trained on 118-bus scenarios

6 Validation Statistics

Table 5: Monte Carlo run statistics.

Parameter	Value
Total LHS scenarios	50,000
Total protection decisions	250,000
Valid decisions	250,000 (0 errors)
Wall time	322 s (5.4 min)
Throughput	≈ 155 scenarios/s
Worker processes	4
Random seed	42
Bus index range	0–117
Fault type distribution	LHS-uniform across {3PH, SLG, DLG, LL}
R_f distribution	Log-normal, $\mu = 0.5$, $\sigma = 1.0$, clip $[0, 20] \Omega$
Load level distribution	Uniform $[0.7, 1.3]$
IBR penetration levels	{0%, 30%, 50%, 100%} (LHS-stratified)
IBR modes	{GFL, GFM, mixed} (LHS-stratified)

6.1 Agree-Rate Confidence Intervals

For the 87L element at 30% IBR penetration, $n = 12,500$ trials with $k = 12,500$ TRIPs gives agree-rate $\hat{p} = 1.000$. The 95% one-sided Wilson lower bound is:

$$p_L = \frac{\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.645, \quad (13)$$

giving $p_L = 99.97\%$, well above the 97% H1 threshold.

7 Conclusions and Contributions

1. **H1 confirmed:** SAMPB 87L achieves 100% agree-rate on the IEEE 118-bus network for IBR penetration $\leq 50\%$, validated over $N = 50,000$ LHS trials spanning all bus locations, fault types, fault resistances, and load levels.
2. **Proportional I_{neg} capping (TR-86-F1):** A new modelling requirement is identified: IBR current limiters must cap I_{neg} proportionally with I_{pos} to maintain the correct symmetrical-component ratio. Failure to do so reduces 87L agree-rate from 100% to 59% at 30% IBR penetration.
3. **100% IBR gap:** Elements 21, 46, and 51 fail H2 at 100% IBR penetration, motivating dedicated redesign in TR-87–TR-89.
4. **Speed:** The 118-bus MC framework processes 155 scenarios/second on 4 CPU cores, enabling rapid what-if studies during relay setting optimisation.

7.1 Agree-Rate Formula — 87L Calibrated Threshold

The final calibrated 87L criterion for IBR networks (applicable for penetration $\leq 50\%$) is:

$$\boxed{\text{TRIP iff } I_{\text{pos}} - 0.25 I_{\text{neg}}^{(\text{cap})} > 0.10 + 0.03 f_{\text{GFM}}} \quad (14)$$

where $I_{\text{neg}}^{(\text{cap})}$ is the proportionally-capped negative-sequence current from Eq. (8).

Source code:

- Network model: `04_code/sampb/digital_twin/models/ieee118_network.py`
- LHS generator: `04_code/sampb/digital_twin/models/mc_scenario_generator.py`
- Protection stack: `04_code/sampb/digital_twin/models/sampb_protection_stack.py`
- MC runner: `04_code/sampb/digital_twin/models/mc_runner_tr86.py`
- Raw results: `04_code/sampb/digital_twin/results/tr86_mc_results.npz`